# Measurement of Planck's and Boltzmann's constants

J. Bergmeier and S. Steinicke

The values $h = (6.51 \pm 0.28) \cdot 10^{-34}$ Js for *Planck's constant* and $k_B = (1.3922 \pm 0.0066) \cdot 10^{-23}$ J/K for the *Boltzmann constant* were found. For the measurement of $h$, we used the fact that it determines the energy of electrons emitted in the *photoelectric effect*. The diffusion current inside an npn-transistor follows a modification of *Schokley's equation* which allowed us to measure $k_B$.

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## 1 Introduction

The idea that natural processes do not need to be continues is quite counterintuitive to most people and hence also baffled physicists at the beginning of the 20th century. But further experimentation showed that the energy quantum introduced by *Max Planck* explained a multitude of open questions better than any other theory. The *Planck constant* $h$, which is inextricably linked to the idea of quantized energy, will be calculated via the *photoelectric effect* in the first part of this paper. Because $h$ connects the frequency and the kinetic energy of free electrons, we will use this experiment to determine this constant.
Besides $h$ another important quantity in quantum mechanics is the *Boltzmann constant* $k_B$ which gives the relation between temperature and the kinetic energy of gas atoms as well as relates the probabilities of different states in one system. $k_B$ is nowadays used to define the SI unit Kelvin, similar to how $h$ defines the Kilogram. In the second part of this paper, we find the *Boltzmann constant* through measurements at a transistor and using the *Shockley diode equation*.

## 2 Planck's constant

An Hg-vapor lamp is used as a source for photons with energy according to *Planck's relation* eq. (1).

$$E = h\nu \tag{1}$$

A lens and aperture are used to focus the light cone on the cathode without hitting the anode of a photocell. Monochromaticity is ensured with different filters that

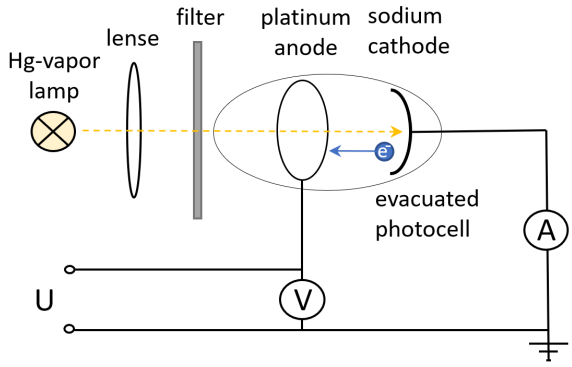


**Figure 1:** The light of an Hg-vapor lamp is shining on a sodium cathode through a lens and filter. Since the photons transfer their energy to electrons in the cathode, the electrons can overcome the exit work and fly to the anode. This photocurrent is measured depending on the voltage, which slows down the electrons.

absorb all undesired parts of the Hg spectrum. Once a light ray hits the sodium cathode, the *photoelectric effect* plays out. The energy $E$ of the photon with frequency $\nu$ is transferred to an electron inside the cathode material. The electron then has enough energy to overcome the exit work $A$ and still continue flying away with kinetic Energy $E_k$ according to *Einstein's equation* eq. (2). Because the energy coming in as light is quantized, the energy of the electron flying away will always depend on the frequency $\nu$ of the incident light, but not on the number of photons. To measure the energy of the electrons, a voltage $U$ is applied that slows the electrons on their way from the cathode to the anode. If the energy $e \cdot U$ lost in the E-field matches the energy of the emitted electrons, the current measured should fall to zero, or at least not continue falling with further increases in voltage. [1]

$$E = h\nu - A = eU \tag{2}$$

The current which is created by the photoelectrons that can reach the anode was measured for different voltages between cathode and anode. This measurement is repeated for different wavelengths $\lambda$: 366 nm, 405 nm, 436 nm, 546 nm, and 578 nm. The characteristic curve for $\nu = 405$ nm is shown in fig. 2, the other curves can be looked up in the appendix fig. 7.

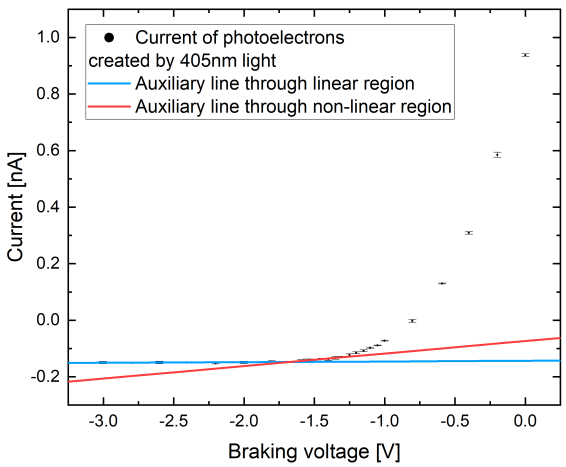


**Figure 2:** The current created by the photoelectrons, that reach the anode is plotted against the voltage between cathode and anode. A blue line was inserted in the linear region of the characteristic curve and a red line was inserted where the points started to deviate from the blue line.

In theory, electrons should not be able to reach the anode at high braking voltages because their kinetic spectrum does not include high enough energies. At the threshold voltage, there should be a clear discontinuity in the characteristic curve because now electrons can reach the anode and create a current. To find the threshold voltage, two lines can be drawn into the graphic. The blue line goes through the values above the threshold (linear region). The reason why the slope here is non-zero is that the exit work is not the same for all cathode atoms because of impurities. Once the measured values begin to deviate from this line, we begin to draw the red line. This line should be tangent to the function through the non-linear region of the characteristic. The voltage at the intersection is then read off. This should be equal to the threshold voltage for the considered frequency. The error was estimated by the range of arguably also valid auxiliary lines. How this was done is shown exemplarily in fig. 2. All the voltages that were found in this way are shown in tab. 1. The wavelengths are used to determine the frequency

| wavelength $\lambda$ | threshold voltage U |
|---|---|
| 366 nm | $(-1.890 \pm 0.078)$V |
| 405 nm | $(-1.67 \pm 0.19)$V |
| 436 nm | $(-1.39 \pm 0.19)$V |
| 546 nm | $(-0.84 \pm 0.19)$V |
| 578 nm | $(-0.63 \pm 0.17)$V |

**Table 1:** graphically determined threshold voltages for different wavelengths

$\nu$ of the light getting past the filter and into the photocell. This was done with eq. (3). For the speed of light $c$, the value 299792458 $\frac{\mathrm{m}}{\mathrm{s}}$ was used.[5]

$$\nu = \frac{c}{\lambda} \tag{3}$$

In fig. 3 the values from tab. 1 were plotted against the matching frequencies. According to eq. (2), the points in this graph are expected to lie on a straight line with slope $s = \frac{h}{e}$.
A linear fit through the points was calculated including the slope $s$ with error. $s$ and its error were then multiplied by the *elementary charge* $e = 1.602176634 \cdot 10^{-19}$ C [5] to obtain *Planck's constant* $h$.

$$h = (6.51 \pm 0.28) \cdot 10^{-34}\ \mathrm{Js}$$

This value overlaps with $6.62607015 \cdot 10^{-34}$Js, the actual value of $h$.[5] Our value is closer to the actual one than the error would lead one to expect. This is not surprising because the method of deriving the graphical error is more in line with an approximation of the maximum deviation rather than the standard deviation.

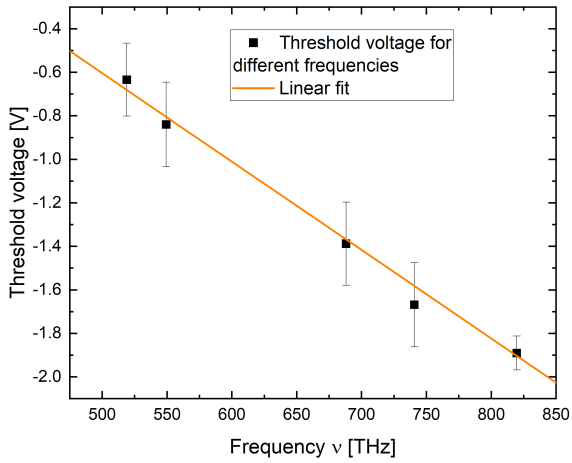


**Figure 3:** The threshold voltages are plotted against the frequency of light with which they were measured. A linear function is fitted through the points.

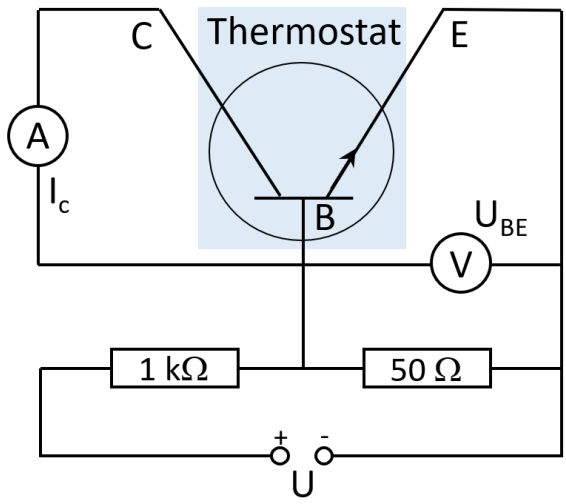


**Figure 4:** Experimental setup to examine the *Boltzmann constant*: The transistor is used to measure the collector current depending on the voltage between basis and emitter. Temperature is controlled by a thermostat.

## 3 Boltzmann's constant

The *Boltzmann constant* can be determined with *Schockley's equation* (eq. (6)). For this purpose, measurements need to be done on a p-n transition. However, due to interfering currents, such as leakage currents at the crystal surfaces, a simple p-n junction is not suitable for measuring the diffusion current. Therefore, a npn-transistor is used. The current is only measured at the collector. Since there is hardly any voltage between the collector and the base, negligible disturbing currents occur there. The circuit diagram is shown in fig. 4. The voltage and current were measured using two *Agilent 34405A* devices. For a more exact adjustment of the voltage between base and emitter, the voltage was tapped at a voltage divided which consisted of two resistors:

$$R_1 = (48.8 \pm 3.8)\,\Omega$$
$$R_2 = (1004 \pm 15)\,\Omega$$

The exact resistor value was checked with a digital multimeter *VC 170-1*, and the associated error was calculated using the data from its manual [4]. The temperature was set with the thermostat *Lauda RE104*. With this, a total of 3 series of measurements on temperatures $T_1 = 40\,°\mathrm{C}$, $T_2 = 80\,°\mathrm{C}$, and $T_3 = 120\,°\mathrm{C}$ were documented.

A potential difference arises at the p-n junction even without external voltage. Two effects lead to a charge carrier movement within the junction. On the one hand, there is the intrinsic conduction current $I_0$ which arises from thermal excitation. The charge carriers are lifted from the valence band into the conduction band and follow the potential gradient. The current relevant for the determination of *Boltzmann's constant* arises from charge carriers diffusing from the positive to the negative region. Some of them have high enough energy to overcome the anti-diffusion voltage. This current is the measured diffusion current $I_D$.

$$\begin{aligned} I_D(U_{BE}) =& C \cdot ek_BT \cdot e^{\frac{e \cdot U_{BE}}{k_BT}} \\ =& |I_0| \cdot e^{\frac{eU}{k_BT}} \end{aligned} \tag{4}$$

eq. (4) stems from the consideration that only particles with energies greater than the counter field contribute to the current. The frequency distribution of the particle energies is given by the Boltzmann statistic (eq. (5)).

$$B = C \cdot e^{\frac{eU_{BE}}{k_BT}} \tag{5}$$

In addition to the *Boltzmann constant*, which we want to determine, the temperature $T$, the base-emitter voltage $U_{BE}$, the elementary charge $e$, and a constant $C$ enter both equations. Since the self-conducting current and the diffusion current cancel each other out when no voltage is applied, the prefactors are equal to the self-conducting current. The total current is described by *Shockley's equation* eq. (6).[1]

$$I(U) = |I_0| \cdot (e^{\frac{eU}{k_BT}} - 1) \tag{6}$$

However, since this experiment uses not only a single p-n junction but a transistor that ensures that at least 95 % of the diffusion current flows uninterrupted, the intrinsic conductance is not important. Thus, to determine the *Boltzmann constant*, eq. (4) is used instead of *Shockley's equation*.

Fig. 5 shows the measured voltage depending on the diffusion current for a temperature T=120°C. The errors were taken from the user manual of the *Agilent*

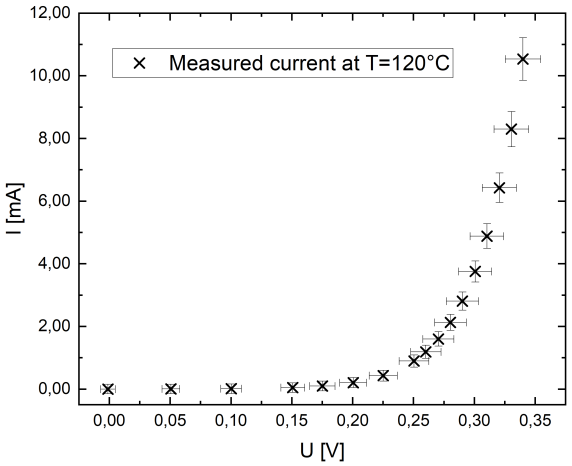


**Figure 5:** Diffusion current measured at the collector as a function of voltage at a temperature of 120°C. The errors are taken from the user manual of the *Agilent 34405A* measuring instruments [2].

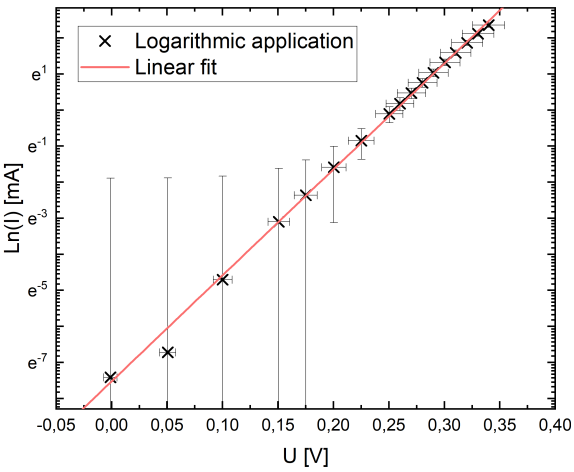


**Figure 6:** Logarithmic scaling of the current to determine the *Boltzmann constant* from the slope of the straight line. This was put through the data with the help of a linear regression, whereby the measured values with very low currents were weighted less.

*34405A* [2].

From the equation for the diffusion current (eq. (4)), the *Boltzmann constant* can be determined by choosing a semi-logarithmic representation. This can be seen in fig. 6. A linear function was fitted through the data with the measured values for small currents weighted less heavily because of the logarithmic plot. The slope $m$ can now be used to calculate the required constant as follows:

$$m = -\frac{e}{k_B T} \leftrightarrow k_B = -\frac{e}{mT} \tag{7}$$

The elementary charge is $e = 1.602176634 \cdot 10^{-19}$ As.[5] Via *Gaussian propagation of uncertainty*, the error was calculated. The required error of the temperature was taken from the manual of the thermostat [3] and the error of the slope from the regression. This procedure was repeated for all 3 measurements. In the appendix, the corresponding graphs can be found. The results are shown in tab. 2.

| Temperature $T$ | Boltzmann constant $k_B$ |
|---|---|
| 40 °C | $(1.398 \pm 0.012) \cdot 10^{-23}$ J/K |
| 80 °C | $(1.393 \pm 0.011) \cdot 10^{-23}$ J/K |
| 120 °C | $(1.386 \pm 0.011) \cdot 10^{-23}$ J/K |

**Table 2:** graphically determined results for the *Boltzmann constant* for 3 temperatures.

From this, an average value can be calculated.

$$k_B = (1.3922 \pm 0.0066) \cdot 10^{-23}\,\mathrm{J/K}$$

Compared to the literature value of the *Boltzmann constant* $k_{B,Lit.} = 1.38064852 \cdot 10^{-23}$ J/K, the determined value deviates by about two sigma bands. It should be noted that the error is very small and assumptions were made in the determination that are not one hundred percent fulfilled in reality. For example, the emitter has a finite resistance that could lead to a systematic error. However, due to the small deviation of 0.8 %, the result obtained is compatible.

# 4 Summary

*Planck's constant* was determined by measuring what voltage is needed to stop electrons produced via the photoelectric effect by photons of known wavelength. A value of $h = (6.51 \pm 0.28) \cdot 10^{-34}$ Js was obtained which matches the actual value of $6.62607015 \cdot 10^{-34}$ Js. In the second section, we determined the *Boltzmann constant* to be $k_B = (1.3922 \pm 0.0066) \cdot 10^{-23}$ J/K. For this, the diffusion current of an npn-transistor depending on the voltage between basis and emitter was

measured for different temperatures. Since the current follows a modification of Shockley's equation, the slope of the associated linear fit could be obtained by a semi-logarithmic plot of the data. The *Boltzmann constant* then was finally calculated from the slope for all three temperatures. Lastly, the mean value was calculated. The deviation from literature value $k_{B,Lit.} = 1.38064852 \cdot 10^{-23}$ J/K is 0.8 %.

## References


[1] Physikalisches Grundpraktikum, Universität Würzburg, Modul C2, Versuch 39, "Plancksches Wirkungsquantum und Boltzmann-Konstante", 2022

[2] User manual *Agilent 34405A*, https://www.manualslib.de/manual/53737/Agilent-Technologies-34405A.html?page=46#manual, last visited: April 7, 2022

[3] Instruction manual *Lauda*, https://www.manualslib.de/manual/135772/Lauda-Re-104.html, last visited: April 7, 2022

[4] Manual of *Voltcraft VC 170-1*, https://www.manualslib.de/products/Voltcraft-Vc170-1-709883.html, last visited: April 7, 2022

[5] National Institute of Standards and Technology, NIST Special Publication 330, 2019 Edition


## 5 Appendix

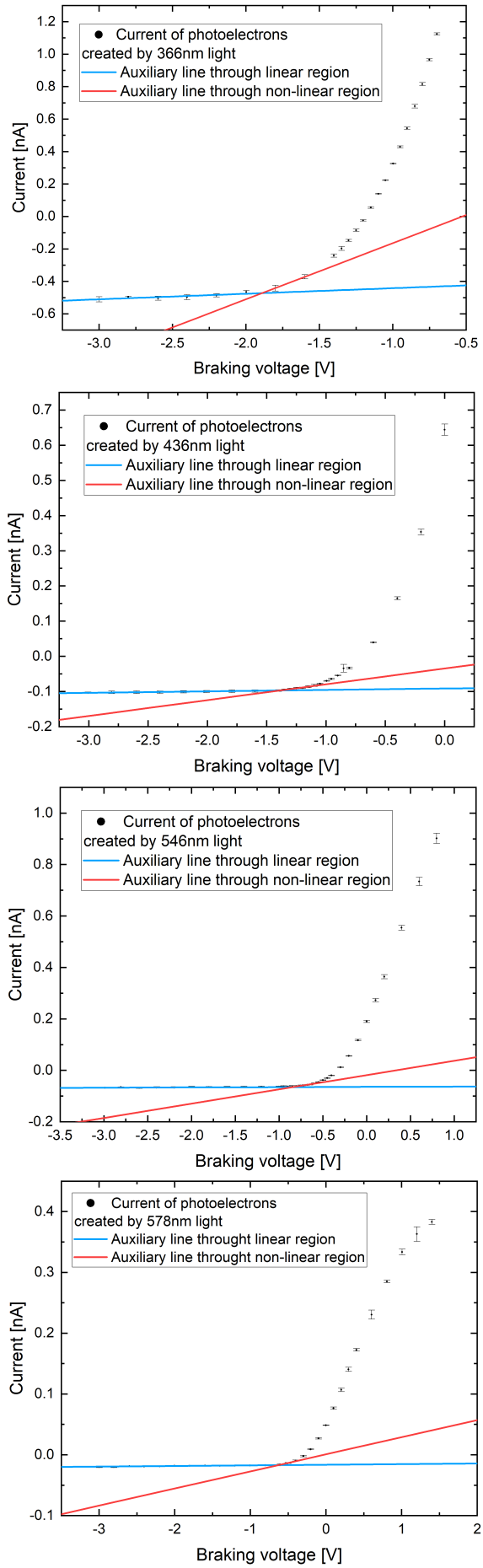


**Figure 7:** Shown are the characteristic curves for all wavelengths not included in the main paper. The blue and red lines show where the threshold voltage was read off from.

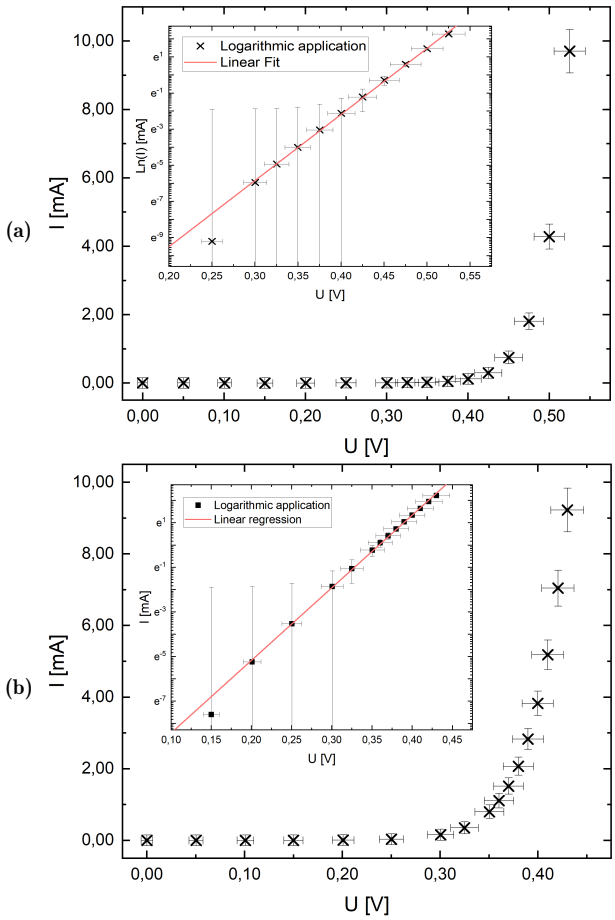


**Figure 8:** Measured current and voltage characteristics, and determination of the *Boltzmann constant* from the slope of the straight line using a logarithmic plot for:
a) T = 40°C
b) T = 80°C